

Kong Konnect ECS Dataplane

Professional Services Asset — Project Assessment Report

Prepared by: Frank Cao | Date: May 08, 2026

Repository: <https://github.com/KongHQ-CX/konnect-ecs-dataplane>

1. Executive Summary

This asset is a **production-ready AWS CDK (TypeScript) library** that automates the deployment of Kong Gateway data planes on AWS ECS Fargate, connecting to **Kong Konnect** as the managed control plane. It is designed as a reusable Professional Services accelerator that customers can adopt with minimal customisation.

The project covers the full operational surface: shared networking (VPC, ALB, WAF), per-service Kong ECS clusters with mTLS to Konnect, control-plane outage resilience via S3, multi-region Route 53 failover, per-service Redis (ElastiCache), centralised log streaming, and a config-file-driven deployment model.

2. Repository Structure

The repository is a single-package CDK application with the following top-level layout:

Asset / File	Purpose
bin/kong-ecs-cdk.ts	CDK app entry point — reads config, instantiates all stacks
lib/kong-infrastructure-stack.ts	Shared infra stack: VPC, ALB, WAF, ACM certificate, Route 53
lib/kong-service-stack.ts	Per-service stack: ECS cluster, Kong DP tasks, ALB listener rule, IAM
lib/constructs/	Seven fine-grained CDK constructs (vpc, alb, waf, certificate, dp-resilience, redis, logging)
config/{env}.json	Environment-specific config file (highest config precedence)
test/	Jest test suite (currently scaffolded; tests to be expanded)

3. Documentation Assets

The repository ships eight Markdown documents. Each addresses a distinct operational concern and is suitable for handing directly to a customer's platform team.

Asset / File	Purpose
--------------	---------

README.md	Project overview, quick-start, full env-var reference, secret format
DEPLOYMENT_GUIDE.md	Step-by-step multi-service deployment, stack outputs, troubleshooting
NAMING_CONVENTIONS.md	Resource naming patterns for shared vs service-specific AWS resources
MULTI_REGION_SETUP.md	Primary / secondary region deployment with Route 53 failover
DP_RESILIENCE_SETUP.md	S3-backed config backup; CP-outage recovery procedure
REDIS_SETUP.md	ElastiCache (Redis) configuration for rate limiting and caching
WAF_CONFIGURATION.md	WAF rule set, CIDR allowlist setup, rate-limit tuning
LOGGING_SETUP.md	CloudWatch log streaming to a central account via subscription filters
architecture-diagram.md	Textual architecture diagram supplement to the PNG

4. Infrastructure Capabilities

Feature	Status	Notes
Multi-service / path routing	Included	Up to 100 services share one ALB; path-priority auto-calculated
mTLS to Konnect CP	Included	Certs stored in Secrets Manager; injected as ECS secrets at runtime
WAF (OWASP + rate limit)	Included	AWS Managed Rules + custom CIDR allowlist; default-block if no CIDRs set
DP resilience (S3 fallback)	Opt-in	Backup ECS node exports config; regular nodes import on CP outage
Multi-region failover	Opt-in	Route 53 health-check failover; set REGIONAL_SUFFIX=secondary
Redis / ElastiCache	Opt-in	Per-service; encryption at rest + in transit; multi-AZ toggle
VPC endpoints	Default on	S3, Secrets Manager, ECR, CloudWatch — no internet traversal
Konnect PrivateLink	Opt-in	VPC endpoint for CP-DP traffic; geo-specific service names
Central log streaming	Opt-in	CloudWatch subscription filters to cross-account Kinesis destination
ARM64 Fargate tasks	Included	All ECS tasks use ARM64 / Linux for cost efficiency
Image via CfnParameter	Included	Container image updatable via CloudFormation without CDK re-run
Force secret refresh	Opt-in	FORCE_REFRESH_TOKEN triggers task restart to pick up new secrets

5. Configuration Model

Configuration is resolved with a three-level precedence chain, making the asset suitable for both local development and CI/CD pipelines:

- **1. config/{environment}.json** — checked into the repo; highest priority
- **2. cdk.context.json / --context flags** — CDK-native context
- **3. Environment variables** — convenient for pipeline injection

Required at minimum:

```
ENVIRONMENT=dev|qa|uat|prd SERVICE1_NAME=customers SERVICE1_PATH=/customers  
SERVICE1_SECRET_ARN=arn:aws:secretsmanager:...
```

Additional services are added by incrementing the index (SERVICE2_*, SERVICE3_*, ... up to SERVICE100_*).